Reverse-check review package — `EffLRSM_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`EffLRSM/repair/final.fr`
original model name	`EffLRSM_gen` (hidden from the agent)
paper	EffLRSM/text/1610.08985.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`LWRQuarksNonHC` (:=)

```
Block[{mu,ii,jj,cc}, -gWRq Sum[CKMR[ii,jj] bar[u[ii,cc]].Ga[mu].ProjP.d[jj,cc] WR[mu],
 ↪  {ii,1,3}, {jj,1,3}, {cc,1,3}]]
```

`LWRQuarks` (:=)

```
LWRQuarksNonHC + HC[LWRQuarksNonHC]
```

`LWRLeptonsNonHC` (:=)

```
Block[{mu,ll,nn}, -gWRl Sum[YR[ll,nn] bar[HN[nn]].Ga[mu].ProjP.l[ll] WR[mu], {ll,1,3},
 ↪  {nn,1,3}]]
```

`LWRLightNuNonHC` (:=)

```
Block[{mu,ll,nn}, -gWRl Sum[XL[ll,nn] bar[ve[nn]].Ga[mu].ProjP.l[ll] WR[mu], {ll,1,3},
 ↪  {nn,1,3}]]
```

`LWRLeptons` (:=)

```
LWRLeptonsNonHC + LWRLightNuNonHC + HC[LWRLeptonsNonHC + LWRLightNuNonHC]
```

`LZRQuarks` (:=)

```
Block[{mu,ii,cc}, gZRq Sum[(bar[u[ii,cc]].Ga[mu].(gZRuL ProjM + gZRuR ProjP).u[ii,cc] +
 ↪  bar[d[ii,cc]].Ga[mu].(gZRdL ProjM + gZRdR ProjP).d[ii,cc]) ZR[mu], {ii,1,3}, {cc,1,3}]]
```

`LZRLeptons` (:=)

```
Block[{mu,ii}, gZRl Sum[(bar[l[ii]].Ga[mu].(gZReL ProjM + gZReR ProjP).l[ii] +
 ↪  bar[ve[ii]].Ga[mu].gZRnuL ProjM.ve[ii] + bar[HN[ii]].Ga[mu].gZRNR ProjP.HN[ii]) ZR[mu],
 ↪  {ii,1,3}]]
```

`LTot` (:=)

```
LWRQuarks + LWRLeptons + LZRQuarks + LZRLeptons
```

Blank-slate reconstruction

Reconstructed Physics Content of `sanitized.fr`

Conventions

Generation indices are $i, j, \ell, n = 1,2,3$, color indices are $\alpha = 1,2,3$. FeynRules `ProjP` and `ProjM` are

$$P_R = \frac{1+\gamma^5}{2}, \qquad P_L = \frac{1-\gamma^5}{2}.$$

`HC[...]` denotes the Hermitian conjugate. The file contains no `DC[...]` covariant derivatives and no `FS[...]` field-strength kinetic terms; only the interaction terms below are defined. The gauge content appearing explicitly is through Standard Model fermion fields u, d, ℓ, ν, color-index contractions for quarks, and new vector bosons $W_R^\pm$, Z_R.

Lagrangian

`LWRQuarksNonHC`

$$\mathcal{L}_{\texttt{LWRQuarksNonHC}} = -\, g_{WRq} \sum_{i,j=1}^{3} \sum_{\alpha=1}^{3} V'_{CKM\,ij}\, \bar{u}_{i\alpha} \gamma^\mu P_R d_{j\alpha}\, W^+_{R\mu}.$$

with

$$g_{WRq} = \kappa_R^q \, \frac{e}{\sqrt{2}\, s_w}.$$

`LWRQuarks`

$$\mathcal{L}_{\texttt{LWRQuarks}} = -\, g_{WRq} \sum_{i,j=1}^{3} \sum_{\alpha=1}^{3} V'_{CKM\,ij}\, \bar{u}_{i\alpha} \gamma^\mu P_R d_{j\alpha}\, W^+_{R\mu} + \text{h.c.}$$

Equivalently, since `CKMR` is declared real in the file,

$$\mathcal{L}_{\texttt{LWRQuarks}} = -\, g_{WRq} \sum_{i,j,\alpha} V'_{CKM\,ij}\, \bar{u}_{i\alpha} \gamma^\mu P_R d_{j\alpha}\, W^+_{R\mu} - g_{WRq} \sum_{i,j,\alpha} V'_{CKM\,ij}\, \bar{d}_{j\alpha} \gamma^\mu P_R u_{i\alpha}\, W^-_{R\mu}.$$

`LWRLeptonsNonHC`

$$\mathcal{L}_{\texttt{LWRLeptonsNonHC}} = -\, g_{WR\ell} \sum_{\ell,n=1}^{3} Y_{\ell N\,\ell n}\, \bar{N}_n \gamma^\mu P_R \ell_\ell\, W^+_{R\mu}.$$

with

$$g_{WR\ell} = \kappa_R^\ell \, \frac{e}{\sqrt{2}\, s_w}.$$

Here N_n is the Majorana fermion class `HN[n]`.

`LWRLightNuNonHC`

$$\mathcal{L}_{\texttt{LWRLightNuNonHC}} = -\, g_{WR\ell} \sum_{\ell,n=1}^{3} X_{\ell\nu\,\ell n}\, \bar{\nu}_n \gamma^\mu P_R \ell_\ell\, W^+_{R\mu}.$$

LWRLeptons

$$\mathcal{L}_{\texttt{LWRLeptons}} = -\,g_{WR\ell}\sum_{\ell,n} Y_{\ell N\,\ell n}\,\bar{N}_n\gamma^\mu P_R \ell_\ell\,W^+_{R\mu} -\,g_{WR\ell}\sum_{\ell,n} X_{\ell\nu\,\ell n}\,\bar{\nu}_n\gamma^\mu P_R \ell_\ell\,W^+_{R\mu} + \text{h.c.}$$

With real `YR` and `XL` as declared,

$$\begin{aligned}\mathcal{L}_{\texttt{LWRLeptons}} = &-\,g_{WR\ell}\sum_{\ell,n} Y_{\ell N\,\ell n}\,\bar{N}_n\gamma^\mu P_R \ell_\ell\,W^+_{R\mu} -\,g_{WR\ell}\sum_{\ell,n} Y_{\ell N\,\ell n}\,\bar{\ell}_\ell\gamma^\mu P_R N_n\,W^-_{R\mu}\\ &-\,g_{WR\ell}\sum_{\ell,n} X_{\ell\nu\,\ell n}\,\bar{\nu}_n\gamma^\mu P_R \ell_\ell\,W^+_{R\mu} -\,g_{WR\ell}\sum_{\ell,n} X_{\ell\nu\,\ell n}\,\bar{\ell}_\ell\gamma^\mu P_R \nu_n\,W^-_{R\mu}.\end{aligned}$$

LZRQuarks

$$\mathcal{L}_{\texttt{LZRQuarks}} = g_{ZRq}\sum_{i=1}^{3}\sum_{\alpha=1}^{3}\left[\bar{u}_{i\alpha}\gamma^\mu\left(g_{ZRuL}P_L + g_{ZRuR}P_R\right)u_{i\alpha} + \bar{d}_{i\alpha}\gamma^\mu\left(g_{ZRdL}P_L + g_{ZRdR}P_R\right)d_{i\alpha}\right]Z_{R\mu}.$$

where

$$g_{ZRq} = -\kappa_R^q\,\frac{e}{s_w}\sqrt{1-\left(\frac{s_w}{c_w\kappa_R^q}\right)^2},$$

$$g_{ZRuL} = -\frac{1}{6}\left(\frac{s_w}{c_w\kappa_R^q}\right)^2,\qquad g_{ZRuR} = \frac{1}{2}-\frac{2}{3}\left(\frac{s_w}{c_w\kappa_R^q}\right)^2,$$

$$g_{ZRdL} = -\frac{1}{6}\left(\frac{s_w}{c_w\kappa_R^q}\right)^2,\qquad g_{ZRdR} = -\frac{1}{2}+\frac{1}{3}\left(\frac{s_w}{c_w\kappa_R^q}\right)^2.$$

LZRLeptons

$$\mathcal{L}_{\texttt{LZRLeptons}} = g_{ZR\ell}\sum_{i=1}^{3}\left[\bar{\ell}_i\gamma^\mu\left(g_{ZReL}P_L + g_{ZReR}P_R\right)\ell_i + g_{ZR\nu L}\,\bar{\nu}_i\gamma^\mu P_L\nu_i + g_{ZRNR}\,\bar{N}_i\gamma^\mu P_R N_i\right]Z_{R\mu}.$$

where

$$g_{ZR\ell} = -\kappa_R^\ell\,\frac{e}{s_w}\sqrt{1-\left(\frac{s_w}{c_w\kappa_R^\ell}\right)^2},$$

$$g_{ZReL} = \frac{1}{2}\left(\frac{s_w}{c_w\kappa_R^\ell}\right)^2,\qquad g_{ZReR} = -\frac{1}{2}+\left(\frac{s_w}{c_w\kappa_R^\ell}\right)^2,$$

$$g_{ZR\nu L} = \frac{1}{2}\left(\frac{s_w}{c_w\kappa_R^\ell}\right)^2,\qquad g_{ZRNR} = \frac{1}{2}.$$

LTot

$$\mathcal{L}_{\texttt{LTot}} = \mathcal{L}_{\texttt{LWRQuarks}} + \mathcal{L}_{\texttt{LWRLeptons}} + \mathcal{L}_{\texttt{LZRQuarks}} + \mathcal{L}_{\texttt{LZRLeptons}}.$$

Field Table

`.fr` class	Particle symbol	Spin	SU(3) rep	SU(2) rep	U(1) charge / hypercharge	Self-conjugate	Mass
`WR`	W_R^+, W_R^-	vector	not declared	not declared	electric charge $Q = +1$ for `WR`, $Q = -1$ for antiparticle	no	$M_{WR} = 3000$
`ZR`	Z_R	vector	not declared	not declared	neutral, inferred from self-conjugacy and interactions; no explicit `Q` entry	yes	$M_{ZR} = 5070$
`HN` / `N1`	N_1	fermion	singlet, no color index	SM $SU(2)_L$ singlet as encoded by absence of weak index and right-chiral couplings	$Q = 0$, inferred from Majorana self-conjugacy	yes, Majorana	$M_{N1} = 173.3$
`HN` / `N2`	N_2	fermion	singlet, no color index	SM $SU(2)_L$ singlet as encoded by absence of weak index and right-chiral couplings	$Q = 0$, inferred from Majorana self-conjugacy	yes, Majorana	$M_{N2} = 1.0 \times 10^{12}$
`HN` / `N3`	N_3	fermion	singlet, no color index	SM $SU(2)_L$ singlet as encoded by absence of weak index and right-chiral couplings	$Q = 0$, inferred from Majorana self-conjugacy	yes, Majorana	$M_{N3} = 1.0 \times 10^{12}$

External Parameters

Parameter	Default value	Appears in	Meaning
`kRquark` / κ_R^q	1	`gWRq`, `gZRq`, `gZRuL`, `gZRuR`, `gZRdL`, `gZRdR`	Overall right-handed quark-sector coupling rescaling relative to e/s_w; controls W_R and Z_R interactions with quarks.
`kRlepton` / κ_R^ℓ	1	`gWRl`, `gZRl`, `gZRnuL`, `gZReL`, `gZReR`	Overall right-handed lepton-sector coupling rescaling relative to e/s_w; controls W_R and Z_R interactions with leptons and heavy neutrinos.
`CKMR[i,j]` / $V'_{CKM\,ij}$	identity matrix	`LWRQuarksNonHC`	Flavor mixing matrix multiplying the charged right-handed quark current $\bar{u}_i \gamma^\mu P_R d_j W^+_{R\mu}$.
`YR[ll,nn]` / $Y_{\ell N\,\ell n}$	identity matrix	`LWRLeptonsNonHC`	Charged-current mixing/coupling matrix between charged leptons ℓ_ℓ, heavy Majorana neutrinos N_n, and $W_R^\pm$.
`XL[ll,nn]` / $X_{\ell\nu\,\ell n}$	zero matrix	`LWRLightNuNonHC`	Charged-current mixing/coupling matrix between charged leptons ℓ_ℓ, light neutrinos ν_n, and $W_R^\pm$.

Physics Summary

The file encodes a simplified right-handed charged and neutral vector sector with a charged vector boson $W_R^\pm$, a neutral vector boson Z_R, and three heavy Majorana neutrinos N_i. The W_R couples to right-chiral quark charged currents and to right-chiral charged-lepton currents involving either heavy Majorana neutrinos or light neutrinos, while the Z_R couples chirally to Standard Model quarks, charged leptons, light neutrinos, and heavy neutrinos. It mediates processes such as $q\bar{q}' \to W_R^* \to \ell N$, $N \to \ell jj$ through off-shell W_R, and neutral-current production or decay through Z_R into fermion pairs including $N_i N_i$.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Located Paper Definitions

The paper defines the model in **Section II, "Effective Left-Right Symmetric Model"**. The relevant definitions are:

- Field content: Section II opening paragraph: SM states, $W_R^\pm$, Z_R, and three heavy Majorana neutrinos N_i.
- Chiral fermion quantum numbers: **Table I**.
- Right-handed charged-current quark interaction: **Eq. (4)**.
- Right-handed charged-current lepton interaction: **Eq. (5)**.
- Heavy/light neutrino mixing assumptions: **Eqs. (6)-(7)**.
- Z_R neutral-current interaction: **Eq. (8)**.
- Z_R chiral coefficients: **Eqs. (9)-(10)**.

- Benchmark mass assumptions: **Eqs. (18)-(19)** and **Table II**.
- Implementation scope: **Section III.A**, which says the SM Lagrangian plus Eqs. (4)-(8) are implemented.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Field content: SM states, $W_R^\pm$, Z_R, three heavy Majorana N_i, with W_R, Z_R aligned to mass eigenstates; N_i aligned with RH chiral states (Sec. II)	Field table lists $W_R^\pm$, Z_R, $N_{1,2,3}$, plus SM fermions appearing in interactions	agree	Reconstruction captures the BSM field content. It does not fully reproduce Table I for all SM chiral fermions, but the interaction terms encode the same charges.
Chiral fermion charges T_L^3, T_R^3, Q_f (Table I)	Uses SM fermions with P_L/P_R and Z_R chiral coefficients inferred from charges	agree	The explicit Table I charge assignments are not reproduced as a full table for all SM fermions, but the reconstructed Z_R coefficients match those charges.
$\mathcal{L}_{W_R qq'} = -\kappa_R^q g/\sqrt{2}\sum \bar{u}_i V_{ij}^{CKM'} W_{R\mu}^+ \gamma^\mu P_R d_j +$ H.c. (Eq. 4)	`LWRQuarks`: $-g_{WRq} V'_{CKMij}\bar{u}_i\gamma^\mu P_R d_j W_{R\mu}^+ +$ h.c., $g_{WRq} = \kappa_R^q e/(\sqrt{2}s_w)$	agree	Since $g = e/s_w$, the coefficient and chirality match. Color contraction is correctly included.
RH CKM matrix taken diagonal with unit entries for the study (text after Eq. 4)	`CKMR[i,j]` default identity matrix	agree	Reconstruction matches the study assumption.
$\mathcal{L}_{W_R \ell\nu/N} = -\kappa_R^\ell g/\sqrt{2}\sum_\ell[\sum_m \nu_m^c X_{\ell m} + \sum_{m'} N_{m'} Y_{\ell m'}] W_{R\mu}^+ \gamma^\mu P_R \ell^- +$ H.c. (Eq. 5) heavy-neutrino part	`LWRLeptonsNonHC`: $-g_{WR\ell} Y_{\ell N}\bar{N}\gamma^\mu P_R \ell\, W_{R\mu}^+$, plus h.c.	agree	Heavy Majorana-neutrino charged current matches coefficient, chirality, and flavor structure.
Eq. (5) light-neutrino part with charge-conjugated light neutrino $\nu_m^c X_{\ell m}$	`LWRLightNuNonHC`: $-g_{WR\ell} X_{\ell\nu}\bar{\nu}\gamma^\mu P_R \ell\, W_{R\mu}^+$	disagree	Reconstruction omits the charge conjugation on the light neutrino. This matters for the field identity/chirality, even though the benchmark sets $X_{\ell m} = 0$.
Mixing scaling $\|Y_{\ell m'}\|^2 \sim O(1)$, $\|X_{\ell m}\|^2 \sim O(m_\nu/m_N)$ (Eq. 6)	External parameters `YR`, `XL`; `YR` identity, `XL` zero by default	agree	Reconstruction encodes the benchmark values, not the scaling statement itself.
Benchmark lepton mixing: Y diagonal with unit entries and all other Y, all X zero (Eq. 7)	`YR` identity matrix, `XL` zero matrix	agree	Assuming generation labels N_1, N_2, N_3, this matches $eN_1, \mu N_2, \tau N_3$.
$\mathcal{L}_{Z_R ff} = -\kappa_R^f g\sqrt{1-(1/\kappa_R^f)^2\tan^2\theta_W}\sum_f \bar{f} Z_{R\mu}\gamma^\mu(g_L^{ZR,f}P_L + g_R^{ZR,f}P_R)f$ (Eq. 8)	`LZRQuarks` and `LZRLeptons` with $g_{ZRq,\ell} = -\kappa_R^{q,\ell} e/s_w\sqrt{1-(s_w/(c_w\kappa_R^{q,\ell}))^2}$	agree	Coefficient structure is equivalent because $g = e/s_w$ and $\tan\theta_W = s_w/c_w$.

paper term (eq. ref)	reconstruction term	verdict	notes
$g_L^{ZR,f} = (T_L^{3,f} - Q_f)\tan^2\theta_W/(\kappa_R^f)^2$ (Eq. 9) for quarks	$g_{ZRuL} = g_{ZRdL} = -\frac{1}{6}(s_w/(c_w\kappa_R^q))^2$	agree	Matches $u_L : 1/2 - 2/3 = -1/6$, $d_L : -1/2 + 1/3 = -1/6$.
Eq. (9) for charged leptons and light neutrinos	$g_{ZReL} = +\frac{1}{2}(s_w/(c_w\kappa_R^\ell))^2$, $g_{ZR\nu L} = +\frac{1}{2}(s_w/(c_w\kappa_R^\ell))^2$	agree	Matches $e_L : -1/2 - (-1) = 1/2$, $\nu_L : 1/2 - 0 = 1/2$.
$g_R^{ZR,f} = T_R^{3,f} - Q_f\tan^2\theta_W/(\kappa_R^f)^2$ (Eq. 10) for quarks	$g_{ZRuR} = 1/2 - \frac{2}{3}(s_w/(c_w\kappa_R^q))^2$, $g_{ZRdR} = -1/2 + \frac{1}{3}(s_w/(c_w\kappa_R^q))^2$	agree	Matches Table I T_R^3 and electric charges.
Eq. (10) for charged leptons and heavy neutrinos	$g_{ZReR} = -1/2 + (s_w/(c_w\kappa_R^\ell))^2$, $g_{ZRNR} = 1/2$	agree	Matches $e_R : T_R^3 = -1/2, Q = -1$ and $N_R : T_R^3 = 1/2, Q = 0$.
Z_R couples to all fermions with $B - L$ charges, including ν_L and N_R (text before Eq. 8)	`LZRLeptons` includes charged leptons, light neutrinos with P_L, and heavy neutrinos with P_R	agree	Reconstruction captures both light and heavy neutral lepton neutral currents.
$M_{Z_R} \simeq 1.7 M_{W_R}$ relation in LRSM but M_{W_R}, M_{Z_R} treated as independent phenomenological parameters (Eq. 18)	Field table gives $M_{WR} = 3000$, $M_{ZR} = 5070$	agree	$5070/3000 \simeq 1.69$, consistent with the quoted representative relation.
Representative masses $M_{W_R} = 3$ TeV, $m_{N_1} = 173.3$ GeV, $m_{N_2}, m_{N_3} = 10^{12}$ GeV (Eq. 19)	Field table gives $M_{WR} = 3000$, $N_1 = 173.3$, $N_{2,3} = 10^{12}$	agree	Matches the benchmark values.
$M_{Z_R} = 5070$ GeV in Table II for representative parameters	Field table gives $M_{ZR} = 5070$	agree	Matches Table II.
SM Lagrangian with Goldstone couplings in Feynman gauge is implemented along with Eqs. (4)-(8) (Sec. III.A)	Reconstruction says the file contains no covariant-derivative or field-strength kinetic terms and only the listed interaction terms	missing-in-reconstruction	If the reconstruction is intended to describe the full paper model, it omits the SM Lagrangian component explicitly mentioned in the implementation section. If it is scoped only to the sanitized BSM interaction file, this is a scope limitation rather than a physics mismatch.
LRSM scalar sector absent/decoupled; non-Abelian W_R/Z_R interactions not correctly modeled in the effective model (Sec. II.B)	No LRSM scalar terms or non-Abelian W_R/Z_R self-interactions reconstructed	agree	The absence is consistent with the paper's stated effective-model limitations.

Disagreements and Checks

1. **Light-neutrino charged-current conjugation** — severity: **substantive**.
   The paper's Eq. (5) uses $\nu_m^c X_{\ell m}$, while the reconstruction writes an unconjugated ν_m; a human should check the original implementation's particle class and FeynRules conventions to see whether the field is actually the charge-conjugated light neutrino or whether the reconstruction normalized it away incorrectly.

2. **SM Lagrangian absent from reconstruction** — severity: **convention** if `reconstruction.md` is intentionally scoped to the sanitized BSM interaction file, otherwise **substantive**.
   The paper's Sec. III.A says the SM Lagrangian is implemented together with Eqs. (4)-(8), so a human should check whether the reconstruction was meant to cover the full UFO/model implementation or only the additional effective LRSM interaction file.

Overall Assessment

The reconstruction matches the paper's effective LRSM interaction structure very closely for the central W_R quark current, heavy-neutrino W_R lepton current, Z_R neutral currents, chiral coefficients, coupling normalizations, benchmark mixings, and benchmark masses. The main physics-level mismatch is the light-neutrino W_R term: the paper writes the light-neutrino contribution with ν^c, while the reconstruction writes an ordinary ν, which affects the field/chirality interpretation even though the benchmark sets that mixing to zero. The only other notable gap is scope-related: the reconstruction does not include the SM Lagrangian that the paper says is implemented, but this may be expected if the unavailable implementation file only contained the extra BSM terms.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 7 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: